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Restoration of circadian behavioural rhythms by gene transfer in *Drosophila*

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The *per* locus of *Drosophila melanogaster* has a fundamental role in the construction or maintenance of a biological clock. Three classes of *per* mutations have been identified: per^l mutants have circadian behavioural rhythms with a 29-h rather than a 24-h period, per^s mutants have short-period rhythms of 19 h, and per^0 mutants have no detectable circadian rhythms[1-4]. Each of these mutations has a corresponding influence on the 55-s periodicity of male courtship song[5]. Long- and short-period circadian rhythm phenotypes can also be obtained by altering the dosage of the wild-type gene[4]: for example, females carrying only one dose of this X-linked gene have circadian rhythms with periodicities about 1 h longer than those carrying two doses. In a previous report[6], cloned DNA was used to localize several chromosomal rearrangement breakpoints that alter *per* locus function. The rearrangements all affected a 7-kilobase (kb) interval that encodes a 4.5-kb poly(A)$^+$ RNA. We report here that when a 7.1-kb fragment from a per^+ fly, including the sequences encoding the 4.5-kb transcript, is introduced into the genome of a per^0 (arrhythmic) fly by P element-mediated transformation, circadian rhythmicity of behaviour such as eclosion and locomotor activity is restored. The transforming DNA complements *per* locus deletions and is transcribed, forming a single 4.5-kb poly(A)$^+$ RNA comparable to that produced by wild-type flies.

The 7.1-kb *Hin*dIII fragment contained in phage ZW106 (represented by a wavy line in Fig. 1) was inserted into Carnegie 20 (ref. 7) to form plasmid pCP1 (Fig. 1). A mixture of pπ 25.1 (ref. 8), which contains a functional P-factor, and pCP1, which contains a defective P-factor that can be rescued by pπ 25.1 and a wild-type *rosy* gene (ry^+) that affects eye colour, was microinjected into the region of pole cell formation of preblastoderm $y\ per^0;ry^{42}$ embryos. Survivors were pair-mated with $y\ per^0;ry^{42}$ flies, and F_1 progeny were scored for wild-type eye colour, which indicates successful germ-line integration of pCP1 sequences (see Fig. 1 legend). Transformed lines were maintained as heterozygotes, with respect to the transforming DNA, by backcrossing to $y\ per^0;ry^{42}$ flies.

One such transformed line, strain per^0;P1.48C ($y\ per^0/Y$; P1.48C/+;ry^{42}/ry^{42} males, and $y\ per^0/y\ per^0$;P1.48C/+; ry^{42}/ry^{42} females), carries a single transposon insertion at polytene chromosomal position 48C (chromosome 2R) by *in situ* hybridization. Hybridization of pCP1 sequences to restriction digests of total genomic DNA from flies carrying P1.48C indicates that the transforming DNA is present in a single copy, and has the structure predicted from the restriction maps in Fig. 1 (data not shown). Derivatives of this strain of flies were subjected to a series of behavioural tests to determine whether transformation of rhythmic behaviour accompanied transformation of eye colour.

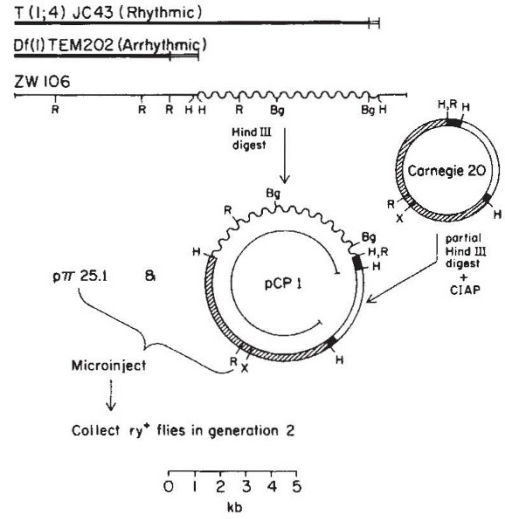


Fig. 1 Physical map of *per* region DNA and the construction of transforming DNA, pCP1. The positions of two chromosomal rearrangements, *T(1;4)JC43* and *Df(1)TEM202*, are shown relative to phage ZW106 (ref. 6). Solid bars represent DNA still present and in the wild-type configuration, while the location of each chromosomal rearrangement breakpoint is shown as an open bar[6] (see text for details).
Methods: Transforming DNA was constructed as follows: ZW106 was digested to completion with the restriction endonuclease *Hin*dIII (H, *Hin*dIII; R, *Eco*RI; Bg, *Bgl*II; X, *Xho*I). The 7.1-kb fragment (shown as a wavy line in ZW106) was purified from a low-melting point agarose gel. Carnegie 20 (ref. 7) was partially digested with *Hin*dIII and treated with calf intestinal alkaline phosphatase (CIAP) to prevent self-religation. The 7.1-kb fragment was ligated to the partially digested vector and introduced into *Escherichia coli* (HB101). Transforming DNA sequences were mapped with restriction endonucleases and one, pCP1 (shown in figure), was chosen for microinjection into *Drosophila*. pCP1 contains the recombinogenic ends of a truncated P-factor[7] (■), *per* locus DNA (〰), a wild-type *rosy* gene (▨), and a combination of DNA sequences from *E. coli* and the *Drosophila melanogaster white* locus[7] (□). The part of pCP1 that is inserted into the *Drosophila* genome by P-factor-mediated transposition includes both per^+ and $rosy^+$ DNA, and is represented by ⊢⊣. pCP1 was mixed with pπ25.1, which mobilizes the P-factor associated with pCP1 in *trans*[8], at a 5:1 concentration ratio (350 μg ml^{-1} final concentration), and the mixture was injected into *Drosophila* embryos of the genotype $y\ per^0;ry^{42}$. Germ-line integration of pCP1 DNA sequences results in a heritable change in eye colour (from *rosy* to wild-type). Because both per^+ and $rosy^+$ DNA sequences are integrated as a unit, flies with ry^+ eye colour were subsequently tested for expression of behavioural rhythms. (For further details, see text.)

Figure 2 shows eclosion profiles for three strains of flies. Two aspects of circadian behaviour were tested: first, the ability of a population of flies to entrain to a light-dark cycle (12 h of light followed by 12 h of dark); and second, the ability of populations of flies to free-run, or continue to behave rhythmically and synchronously, in the absence of the entraining light-dark cycle. In the experiments of Fig. 2, flies were kept in constant darkness from day 6 until the end of the experiment.

Figure 2*a* shows that flies carrying one dose of a wild-type (Oregon-R) *per* locus ($y\ per^0/+;ry^{42}/+$females) entrain to a light-dark cycle and continue to show circadian eclosion rhythms in constant darkness. The period of these rhythms is ~24 h during entrainment, and 25 h in constant darkness (see Fig. 2 legend). As expected[1-3], per^0 flies ($y\ per^0/Y;ry^{42}/ry^{42}$ males, and $y\ per^0/y\ per^0;ry^{42}/ry^{42}$ females) eclose arrhythmically in both sets of conditions (Fig. 2*b*). Flies in the population designated per^0:P1.48C (Fig. 2*c*) differ from those in the per^0 population only by the presence of the transforming DNA on one of their second chromosomes, at 48C ($y\ per^0/Y$;P1.48C/+;

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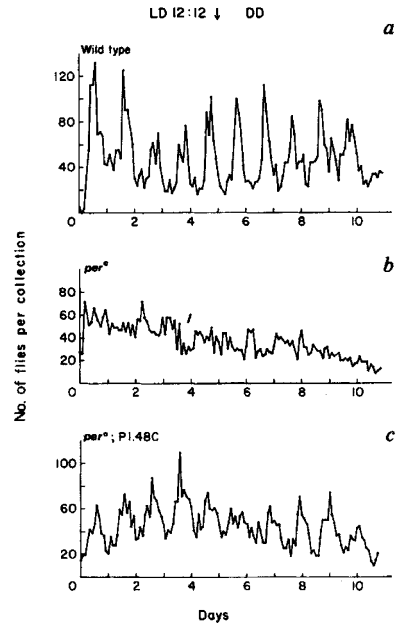


Fig. 2 Temporal profiles of eclosion (emergence of adults from pupal cases) for populations of: *a*, wild-type females ($y\ per^0/+$; $ry^{42}/+$); *b*, per^0 males and females ($y\ per^0;ry^{42}/ry^{42}$); and *c*, per^0;P1.48C males and females ($y\ per^0;P1.48C/+;ry^{42}/ry^{42}$). Eclosion was monitored simultaneously in the three different populations by manually collecting (at 2-h intervals) the adults that had emerged in culture bottles maintained at 25 °C. Developing cultures were entrained to a cycle of 12 h light/12 h dark for 3 days before the start of collections. This light-dark (LD) cycle was continued for the first 5 days of collections, then cultures were maintained in constant darkness (DD) for the remainder of the experiment. The LD to DD transition is indicated by the arrow above record *a*. Collections during DD were made under a low-intensity safelight (a 15 W incandescent light with a Kodak GBX-2 filter) which does not affect *Drosophila* circadian rhythms[1,2]. During entrainment to LD 12:12, 24-h period lengths were estimated for the rhythms of wild-type (*a*) and per^0;P1.48C (*c*) populations by measuring the time elapsed between medians of successive peaks of eclosion. By the same procedure, wild-type and per^0;P1.48C populations have free-running period lengths (in DD) of 25 ± 1.1 h and 26.5 ± 1.9 h, respectively. Similar estimates (24–25 h for wild-type and 27–28 h for per^0;P1.48C) were obtained from a spectral analysis of the data (Univariate Spectral Analysis BMDP P1T, University of California, Los Angeles; see ref. 11 for methods). Spectral analysis revealed no periodicity in the eclosion profile for per^0 flies.

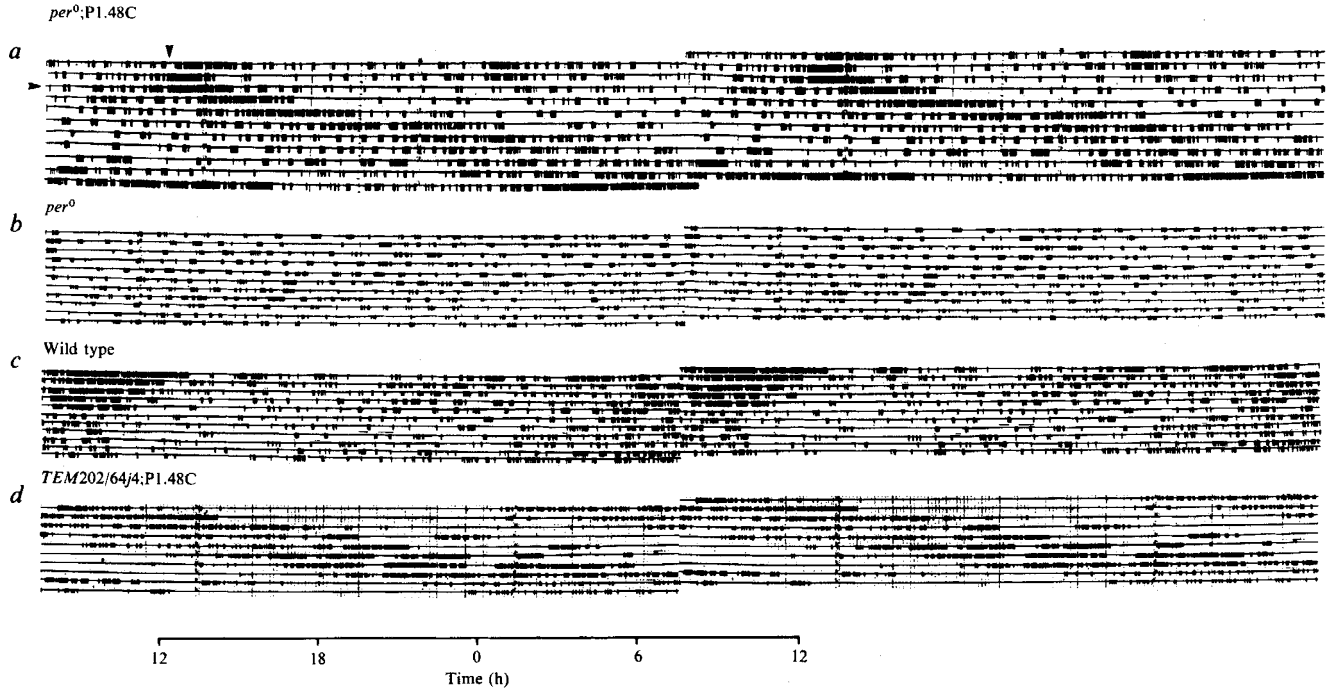


Fig. 3 Records of locomotor activity from individual flies: *a*, $y\ per^0$/Y;P1.48C/+;$ry^{42}/ry42$ male; *b*, $y\ per^0/y\ per^0;ry^{42}/ry^{42}$ female; *c*, Oregon-R wild-type male; and *d*, *Df(1)TEM202/Df(1)64j4*;P1.48C/+female. Data were collected by maintaining individuals, at 25 °C, in cylindrical plastic tubes (containing a small amount of medium) positioned between a light-emitting diode and a phototransistor of the monitoring device[12]. The diode used emits light in a range (peak emission 900 nm) that does not affect circadian rhythms of locomotor activity[12]. When a fly's movement disturbed the light beam, an all-or-none signal was generated which, when amplified, triggered a pen deflection on an Esterline Angus event recorder. Records were constructed by plotting successive days (24 h) of activity one below the other, then plotting twice the resulting chart of activity[13]. Thus, each record is shown twice, but the second record in every pair is displaced upwards 1 day on the right so that a continuous 48-h interval is shown on each horizontal line. In *a*, the first 3 days of the record show entrainment of the per^0;P1.48C fly to a cycle of 12 h light/12 h dark (LD 12:12). Lights were switched on at 0 and off at 12 h. The final 'lights off' signal of the LD cycle is indicated by the intersection of the two arrows (12.00 on day 3). Thereafter, the fly remained in continuous darkness. For *b*, *c* and *d*, only activity in constant darkness is plotted. Period length was determined by measuring the time elapsed between offsets (cessations) of daily bouts of activity[1]. All flies were entrained to LD 12:12, with 'lights on' and 'lights off' at the times indicated above, for at least 2 days before each activity recording began.

ry^{42}/ry^{42} males, and $y\ per^0/y\ per^0$;P1.48C/+;ry^{42}/ry^{42} females). In sharp contrast to per^0 flies, per^0;P1.48C flies eclose rhythmically during entrainment and continue to eclose rhythmically in constant darkness (Fig. 2*c*). The period of eclosion rhythms during entrainment is 24 h, whereas the period of free-running rhythms is ~27 h (see Fig. 2 legend).

Representative locomotor activity records for individual per^0;P1.48C, per^0 and Oregon-R flies are shown in Fig. 3; each horizontal line of each record represents 48 h. Individual per^0;P1.48C adults entrain to a light-dark cycle and continue to behave rhythmically in constant darkness (Fig. 3*a*). The first three days of the activity record in Fig. 3*a* ($y\ per^0$/Y;P1.48C/+; ry^{42}/ry^{42} male) illustrate entrainment to a cycle of 12 h of light followed by 12 h of dark. Bursts of locomotor activity occur at dawn and dusk. The light-dark cycle ends on day 3 at 12.00. Bouts of activity in constant darkness continue to occur rhythmi-

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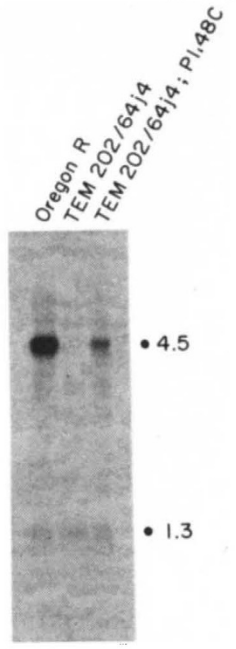


Fig. 4 Transcriptional activity of P1.48C. Poly(A)$^+$ RNAs from flies of the genotypes shown (5 μg per lane) were electrophoresed through 1% agarose/2.2 M formaldehyde gels and transferred to Biodyne A membrane (Pall Ultrafine Filtration). The 7.1-kb *Hin*dIII fragment from phage ZW106 (Fig. 1) was cloned into the Riboprobe vector pSP64 (Promega Biotec). The autoradiogram shows results from the hybridization of a single-stranded, ^{32}P-labelled, 1.5-kb RNA probe. The probe was initiated at the SP6 promoter of the vector and terminates at a *Bst*EII site within the 7.1-kb *Hin*dIII fragment; this *Bst*EII site is located 1 kb to the left of the rightmost *Bgl*II site shown in the ZW106 map of Fig. 1. This probe labels two RNAs after washing at 65 °C in 5 mM NaCl, 2 mM $NaPO_4$ *p*H 7, 0.1 mM EDTA, 0.1% SDS. The 1.3-kb RNA (indicated) is present in all samples, including flies deficient for 10 kb of DNA lying between the breakpoints of *Df(1)TEM202* and *Df(1)64j4*. As these flies lack the 7.1-kb *Hin*dIII fragment, the 1.3-kb RNA cannot be transcribed from this DNA[6]. The 4.5-kb RNA is absent from these flies, but is present in wild-type flies (Oregon-R), and is found in reduced amounts in flies of the genotype *Df(1)TEM202/Df(1)64j4*; P1.46C/+. A densitometric analysis of shorter exposures of the autoradiograph indicates a 10-fold difference in the level of the 4.5-kb RNA between *Df(1)TEM202/Df(1)64j4*; P1.48C/+ and wild-type flies; a fivefold reduction from wild type would be expected in flies carrying two copies of P1.48C. Because the hybridization probe in these experiments is single-stranded, the direction of the transcription of the 4.5-kb RNA is from left to right along the 7.1-kb *Hin*dIII fragment shown in Fig. 1. Hybridizations using labelled RNA probes synthesized from the 7.1-kb *Hin*dIII fragment cloned in the opposite orientation fail to detect any homologous RNA species in the three genotypes shown.

cally with a period of ~27 h; each bout of activity occurs ~3 h later than that of the preceding day. Comparable results were obtained with 7 of 8 additional *per*^{0};P1.48C transformants, each monitored for a minimum of five free-running daily cycles (mean period = 28.1 ± 1.2 h). Thus, both eclosion rhythms and locomotor activity rhythms have approximately the same period.

Figure 3*b* shows the locomotor activity of a *per*^{0} fly (*y per*^{0}/*y per*^{0};*ry*^{42}/*ry*^{42} female) in constant darkness. As with the eclosion behaviour of *per*^{0} populations, no rhythmic activity can be seen. Figure 3*c* shows a representative locomotor activity record of a free-running Oregon-R fly (male); the period of the rhythm in constant darkness is slightly less than 24 h. The period of the rhythms of such euploid, Oregon-R males is about an hour shorter than that of females carrying a single dose of *per*$^+$ (Fig. 2); this result is expected because *per* is X-linked and dosage-compensated[4], and period length is sensitive to *per* locus dosage[4].

Df(1)TEM202/Df(1)64j4 females are arrythmic and homozygously deficient for a 10-kb interval that includes the 7.1-kb *Hin*dIII fragment[6]. To determine whether the transforming DNA sequence in P1.48C can genetically complement this *per* locus deletion, a new group of flies (*Df(1)TEM202/Df(1)64j4*; P1.48C/+females) was constructed. Figure 3*d* shows a representative locomotor activity record for a free-running *Df(1)TEM202/Df(1)64j4*;P1.48C fly; the fly is rhythmic, the period of the rhythm in constant darkness being 27 h. Of 25 flies tested, 24 produced comparable rhythms, with a mean period of 27.6 ± 2.4 h. No measurable difference could be found between the rhythms of *per*^{0};P1.48C and *Df(1)TEM202/Df(1)64j4*;P1.48C flies.

A 4.5-kb poly(A)$^+$ RNA is transcribed from the 7.1-kb *Hin*dIII fragment in wild-type flies[6]. To determine whether the P1.48C transforming DNA produces this RNA, two lines of *Df(1)TEM202/Df(1)64j4* aneuploid flies were constructed, one carrying the transposon insertion (that is, *Df(1)TEM202/Df(1)64j4*; P1.48C/+). Figure 4 shows that the 4.5-kb RNA is expressed in aneuploid flies only if P1.48C is present; it also demonstrates that the 4.5-kb RNA is much more abundant in wild-type than in *Df(1)TEM202/Df(1)64j4*; P1.48C flies (see Fig. 4 legend also).

Our analysis of *Drosophila* strain *per*^{0};P1.48C proves that a 7.1-kb *Hin*dIII fragment derived from a wild-type fly contains a functional *per* locus, and behavioural analysis of *Df(1)TEM202/ Df(1)64j4*;P1.48C flies shows that the transforming DNA can complement *per* locus deletions. *Df(1)TEM202/Df(1)64j4*;P1.48C flies are still completely deficient for two transcription units neighbouring the 7.1-kb *Hin*dIII fragment; these map near the *T(1;4)JC43* chromosomal rearrangement breakpoint and code for a 1.1- and a 3.2-kb poly(A)$^+$ RNA[6]. Because *Df(1)TEM202/Df(1)64j4*;P1.48C flies are rhythmic, it can be concluded that neither transcript has an essential role in the production of circadian behavioural rhythms.

The 7.1-kb *Hin*dIII fragment was chosen for these transformation experiments because it appeared to contain a single transcription unit that forms a 4.5-kb RNA[6]. As the 4.5-kb poly(A)$^+$ RNA is in fact produced by the P1.48C transposon insertion, this RNA appears to be required for expression of behavioural rhythms.

As judged by an analysis of eclosion and locomotor activity rhythms, P1.48C restores the ability of *per*^{0} flies to entrain to light–dark cycles and to express rhythmic behaviour in constant darkness. The only measurable difference between the rhythms of wild-type and *per*^{0};P1.48C flies is the length of the free-running period. The finding that P1.48C transformants have long-period free-running rhythms is not particularly surprising given that period length is known to be sensitive to *per* locus dosage, with a 50% reduction in wild-type gene dosage increasing free-running rhythmic period length by 1 h. Because the level of expression of transforming DNA can be strongly influenced by chromosomal location in *Drosophila*[9,10], we suspect that P1.48C produces less than wild-type levels of a functional *per* gene product. Consistent with this interpretation, we find that *Df(1)TEM202/Df(1)64j4*;P1.48C/+ flies produce 10 times less than the wild-type level of 4.5-kb RNA. An analysis of the relationship of rhythmic period length and abundance of the 4.5-kb RNA among several strains of flies independently transformed with the 7.1-kb *Hin*dIII fragment is under way.

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Note added in proof: A second *Drosophila* strain independently transformed with the 7.1-kb *Hin*dIII fragment, produces rhythms with close to wild-type periodicity. Of 15 such flies tested, 14 produced locomotor activity rhythms with a period 1 h shorter than flies transformed by P1.48C. A molecular analysis of *per* (ref. 6) has also been initiated by Reddy *et al.*[14].

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